

SEQUENTIAL LOGIC DESIGN

Module.3

CDA 3103C

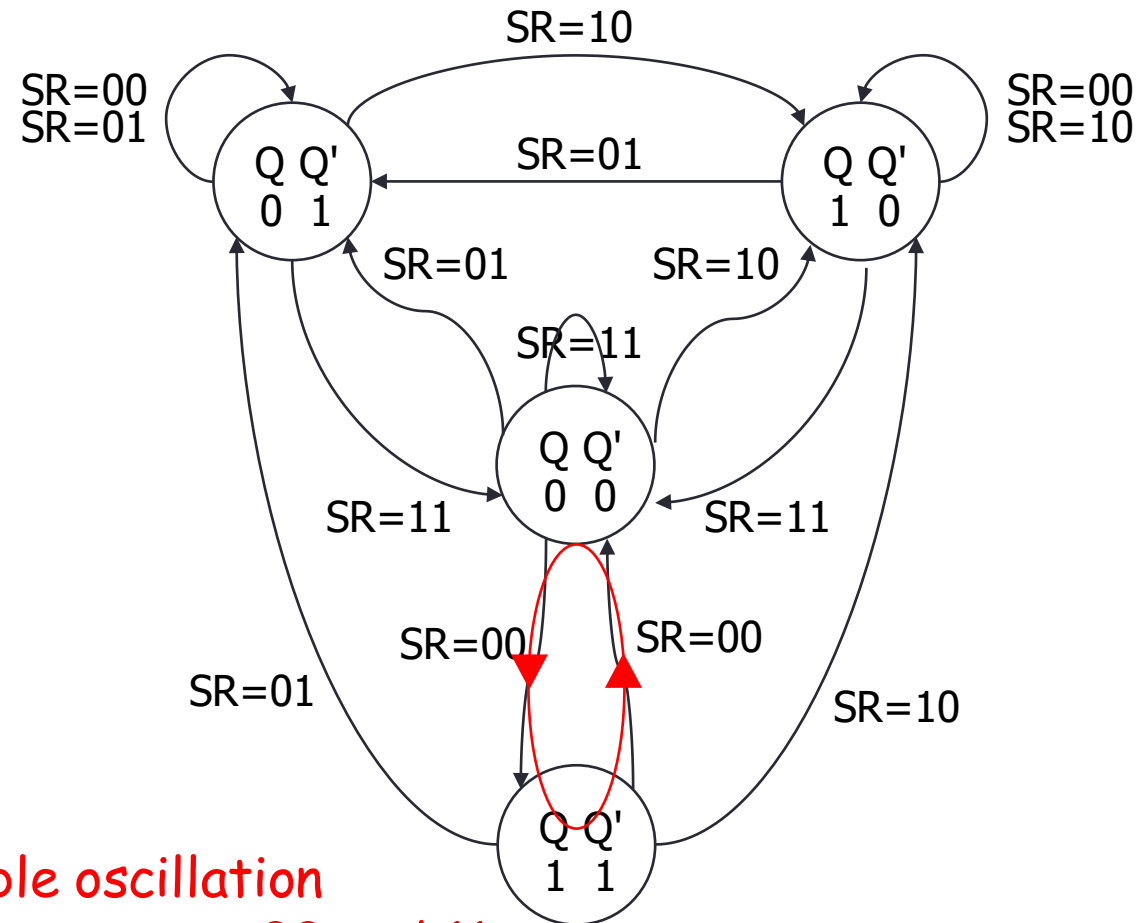
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Behavior of the S-R Latch

- State Diagram
 - States: possible outputs
 - Transitions: events that cause the system to change states (inputs)

S	R	State
0	0	hold
0	1	0
1	0	1
1	1	unstable

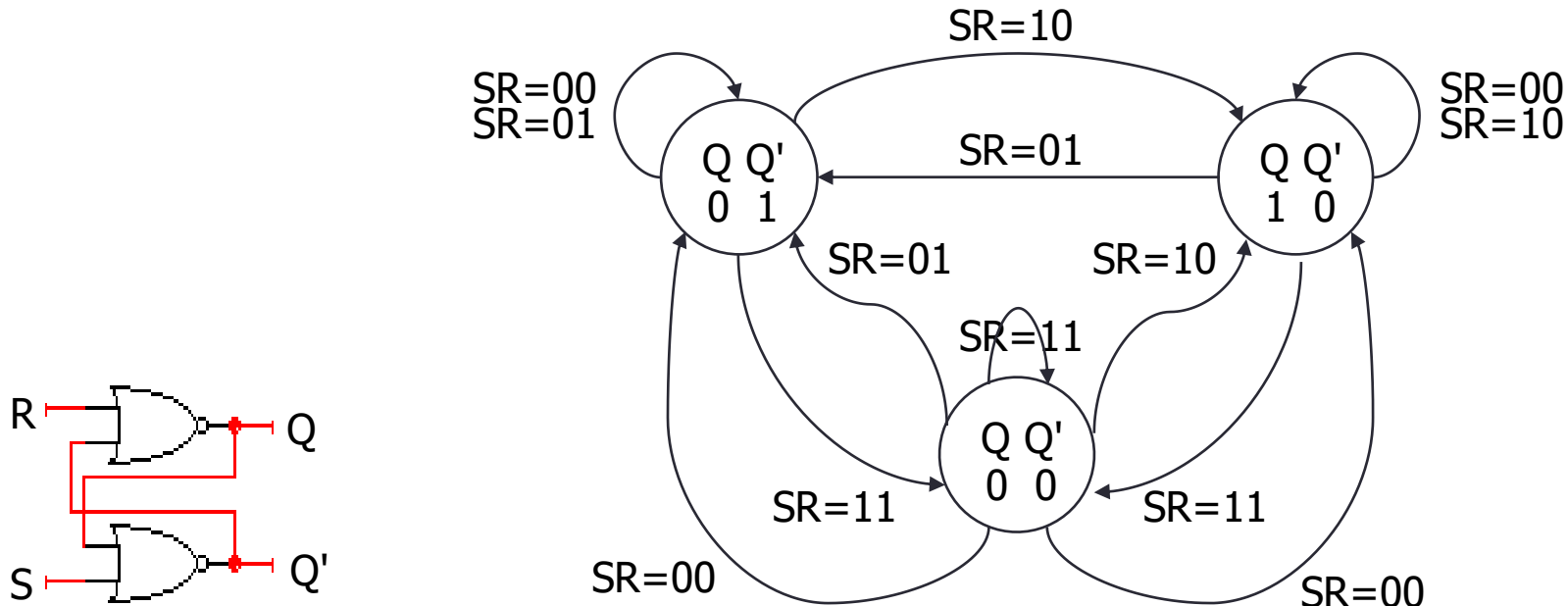
S=0, R=0, Q=0, Q'=1
S=1, R=0, Q=1, Q'=0
S=0, R=0, Q=1, Q'=0
S=0, R=1, Q=0, Q'=1
S=1, R=1, Q=0, Q'=0



possible oscillation
between states 00 and 11

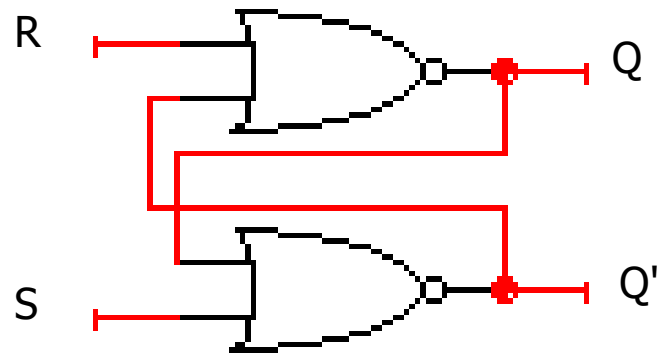
Behavior of the S-R Latch

- Very difficult to observe R-S latch in the unstable state
 - One of R or S usually changes first
- Ambiguously returns to state 0-1 or 1-0
 - A so-called "race condition"
 - Or non-deterministic transition



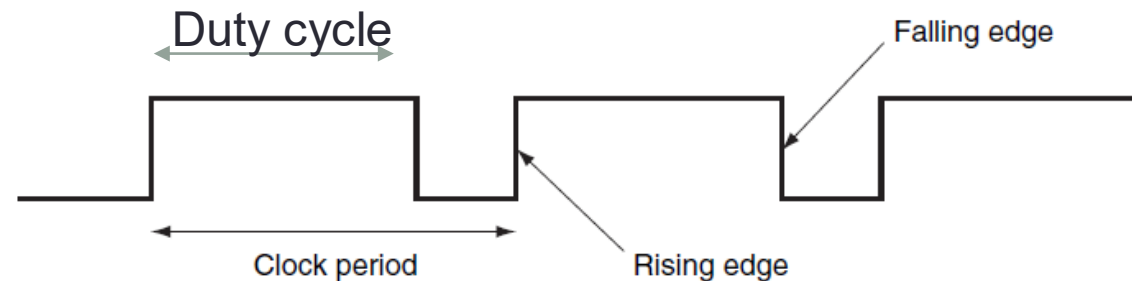
Unclocked S-R Latch

- In an unclocked circuit, the outputs change whenever the inputs change.



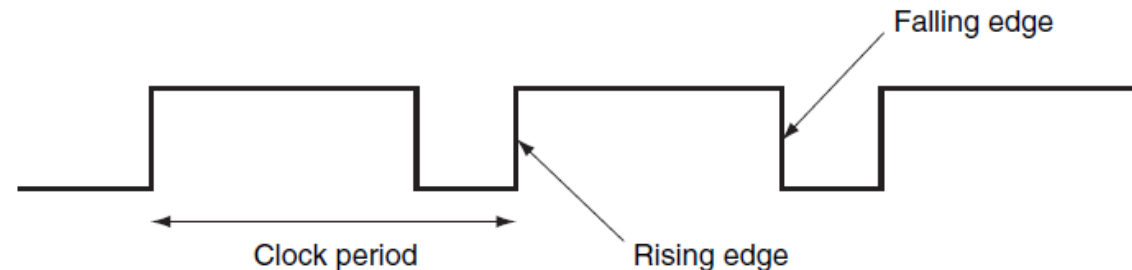
Clocks

- Clocks are regular periodic signals
 - Clock period is divided into two portions
 - When the clock is “high”, also called the duty cycle
 - When the clock is “low”
 - State changes occur on a clock edge
 - When the clock moves from low to high or from high to low



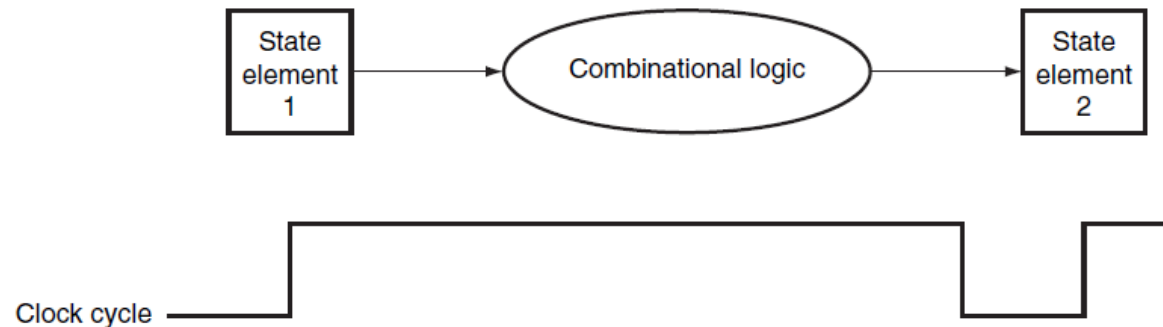
Edge-Triggered Methodology

- Either the rising edge or the falling edge is active
 - Determined by implementation
 - Does not affect design
- Elements change on the active clock edge
 - All inputs sampled at the same time
 - All inputs must be valid (steady) at the active clock edge
- Clocked systems are also called synchronous systems.



Sequential Logic Design

- Combinational logic elements are still used
- State elements provide valid inputs to the combinational logic block.
- Clock period must be long enough to allow all the signals in the combinational logic block to settle

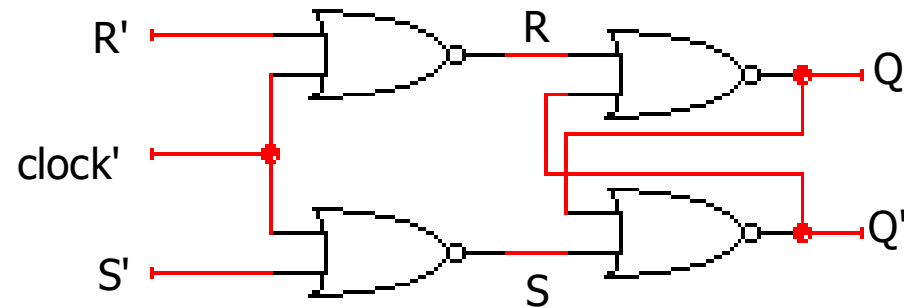


Steady-State Abstraction

- All real circuits need time to update
 - The outputs do not change instantaneously after an input change
- A fundamental abstraction of digital design is to reason about steady-state behaviors
 - Look at outputs only after sufficient time has elapsed for the system to make its required changes and settle down
 - Memory of a system is represented as its state
 - Changes in system state are only allowed to occur at specific times

Clocked S-R Latch

- In a clocked circuit, we control when the inputs are sampled.

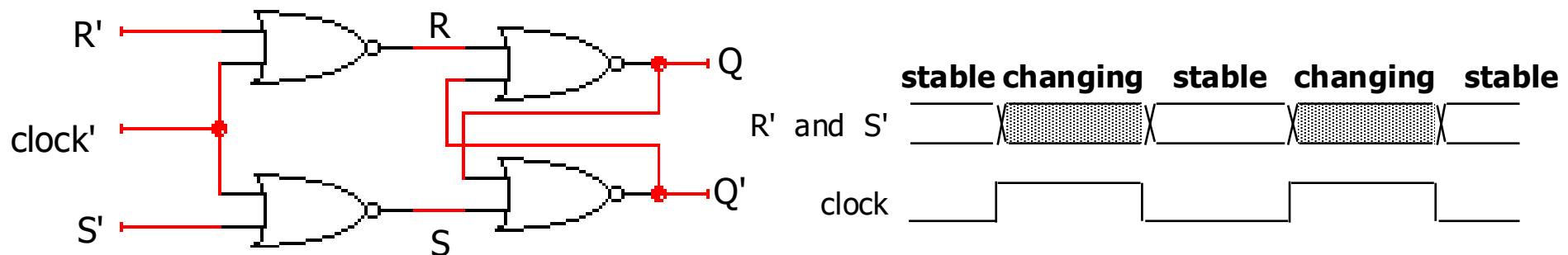


Demo:

<https://www.youtube.com/watch?v=VPqOExHQd4M>

Clocked S-R Latch

- Controlling an R-S latch with a clock
 - Can't let R and S change while clock is active
 - Only have half of clock period for signal changes to propagate
 - Signals must be stable for the other half of clock period

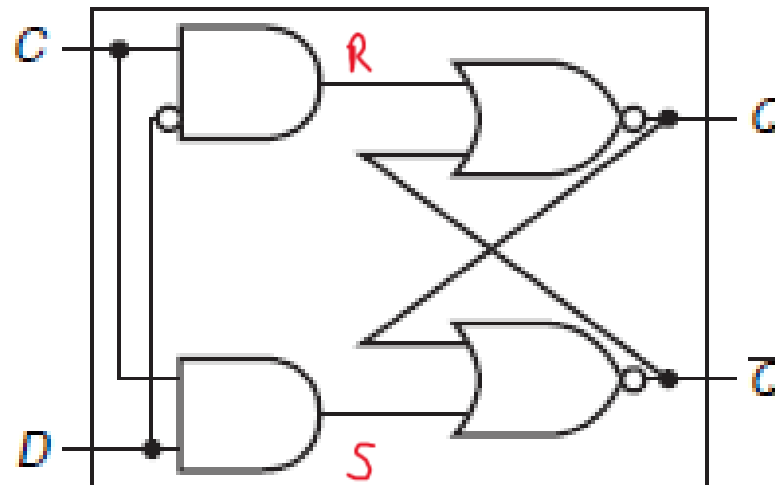


Flip-Flops

- A flip-flop is a memory element for which the output is equal to the value of the stored state inside the element and for which the internal state is changed only on a clock edge.
 - Often built from clocked latches.

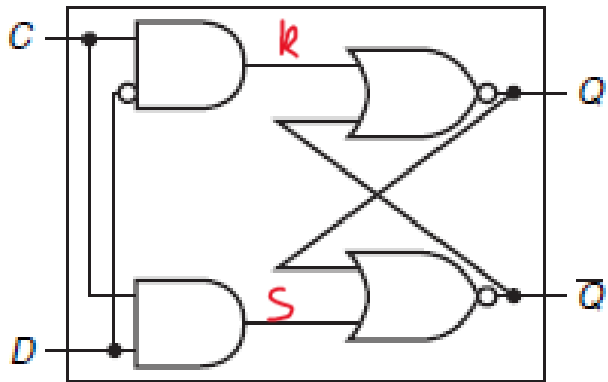
D Flip-Flop

- 2 Inputs
 - D, the data to be stored
 - C, the clock signal
- 2 Outputs
 - Q, the internal state
 - Q', the complement of Q



D Flip-Flop

- When the clock is asserted, the flip-flop is “open”
 - If D is 1, the S-R latch is “set”
 - If D is 0, the S-R latch is “reset”
- When the clock is deasserted, the flip-flop is “closed”
 - The S-R latch is “holding”

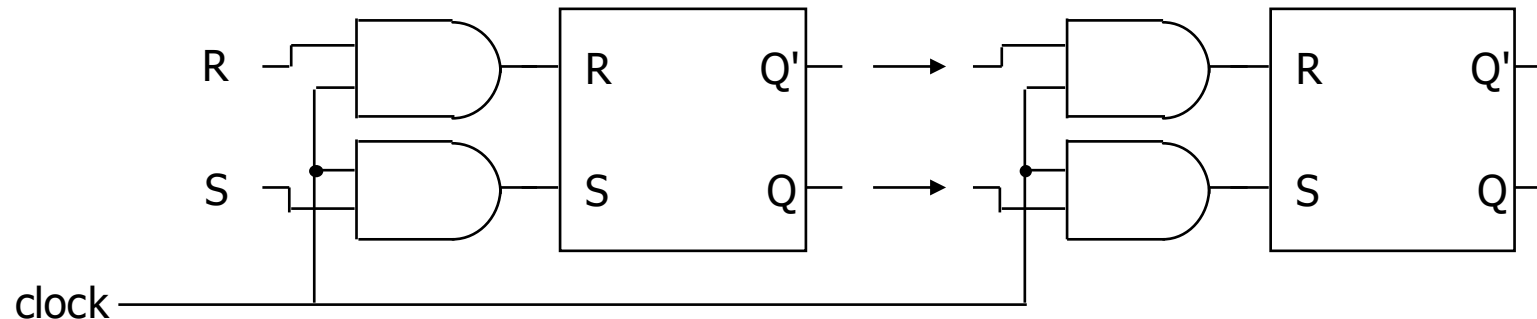


D Flip-Flop

- Flip-Flops can be built to trigger on either the rising or falling clock edge
 - Remember this does not affect the design
- Flip-Flops can be built to trigger on the offset edge
 - If all Flip-Flops trigger on the positive edge, we can build one to trigger on the negative edge
 - We use cascading logic blocks

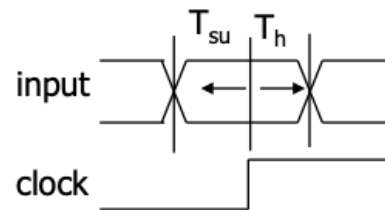
Cascading Logic Circuits

- Connect output of one latch to input of another
- How to stop changes from racing through chain?
 - Need to control flow of data from one latch to the next
 - Move one latch per clock period
 - Must respect setup and hold time constraints to successfully capture input

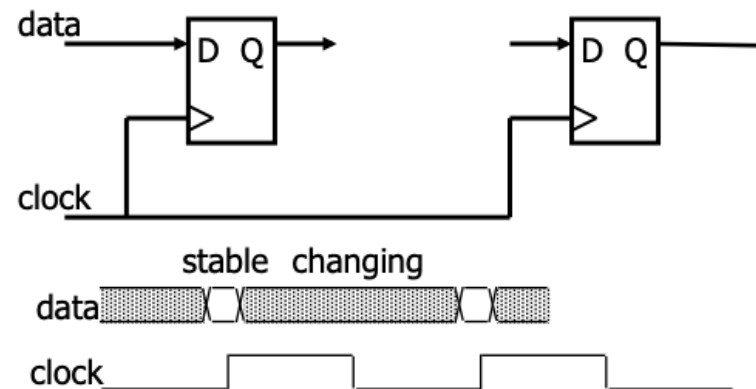


Timing Methodologies

- **Clock**: periodic event, causes state of memory element to change can be rising edge or falling edge or high level or low level.
- The **setup time** is the minimum time that the input must be valid before the clock edge. (T_{setup} or T_{su})
- The **hold time** is the minimum time during which it must be valid after the clock edge. (T_{hold} or T_{h})
- Metastability: the output going into unresolved state.



there is a timing "window" around the clocking event during which the input must remain stable and unchanged in order to be recognized

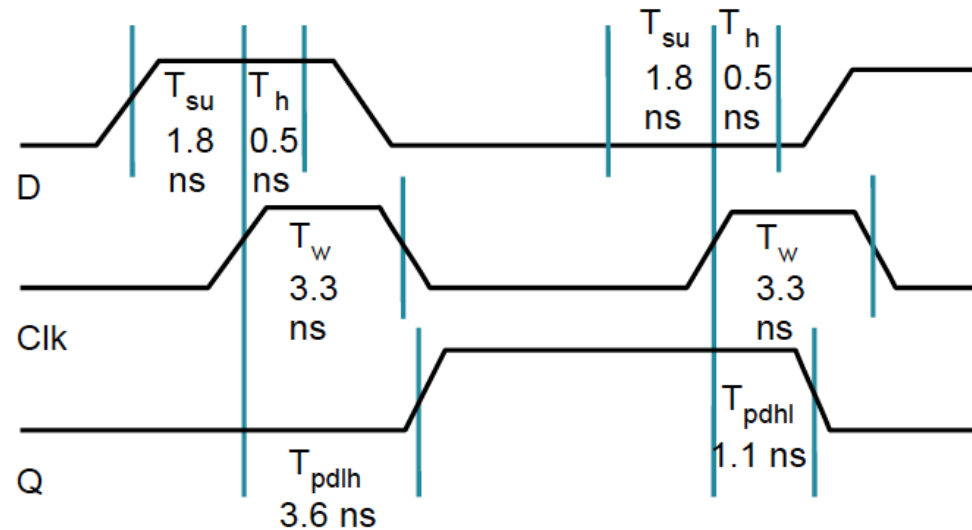


Timing Methodologies

- Rules for interconnecting components and clocks
 - Guarantee proper operation of system when strictly followed
- Approach depends on building blocks used for memory elements
 - Focus on systems with edge-triggered flip-flops
- Basic rules for correct timing:
 - (1) Correct inputs, with respect to time, are provided to the flip-flops
 - (2) No flip-flop changes state more than once per clocking event

Typical Timing Specifications

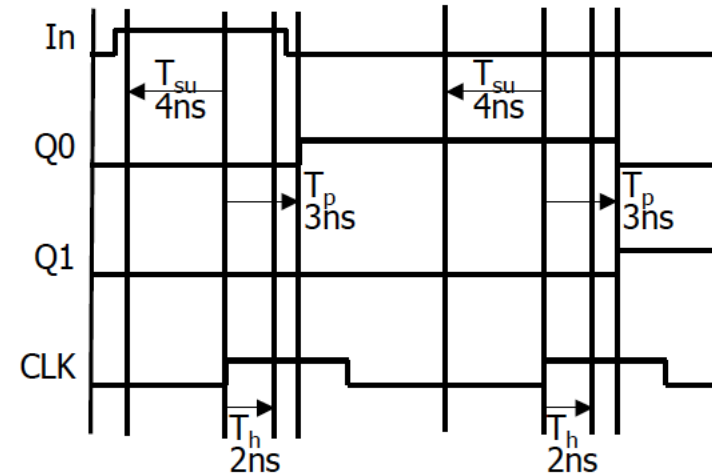
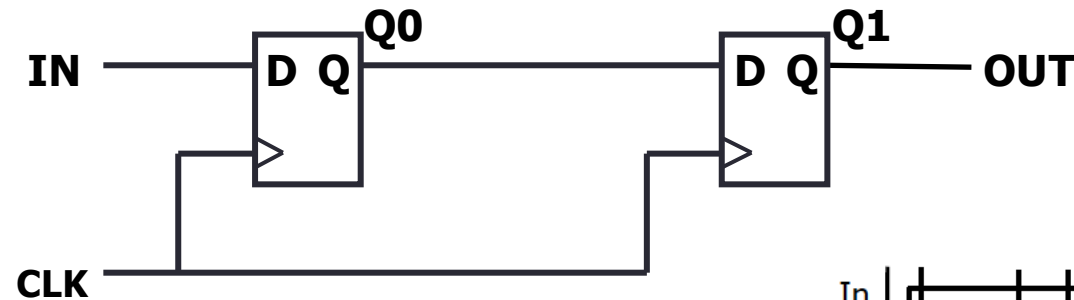
- Positive edge-triggered D flip-flop
 - Setup and hold times
 - Minimum clock width
 - Propagation delays (low to high, high to low)



all measurements are made from the clocking event:
the rising edge of the clock

Cascading Edge-triggered Flip-Flops

- Consider setup/hold/propagation delays
 - prop must be $>$ hold





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